

TITLE PAGE

- **Problem Statement ID** - 26113
Problem Statement Title - Accessibility and quality of public healthcare services, particularly in rural and underserved areas
Theme - MedTech / BioTech/Health Tech
PS Category - Software
Team ID - [Blank/To be filled by team]
Team Name (Registered on portal) - The Persimmons



Project MedSahayak

- **Detailed explanation of the proposed solution**

A health website that turns confusing medical information into something anyone can understand, by providing plain-language explanations of medical as medical reports, prescriptions, and X-rays, and consolidating all health records. It tracks health over time with simple graphs.

- **How it addresses the problem**

By explaining jargon and consolidating records, it directly improves the 'accessibility' and 'quality' of services for users (especially in underserved areas) who may struggle with medical terminology. Features like AI help, voice-first UI (later pages), and reminders specifically target rural and elderly demographics.

- **Innovation and uniqueness of the solution**

Unlike other tools, it's built around 'understanding', not just storage. Every report is accompanied by a plain-language explanation. The AI support is always available to clarify any queries.

TECHNICAL APPROACH

Technical Approach

Methodology

Capture

Patient photographs or scans a medical report, prescription, or X-ray using basic smartphone.

Simplify

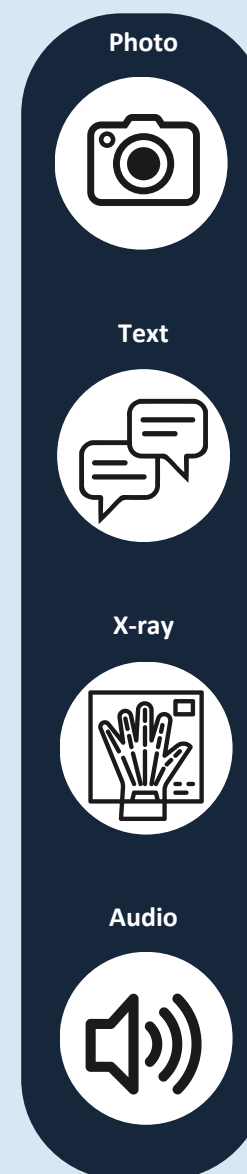
OCR + NLP turn medical jargon into plain-language explanations – what it says and what medicine means.

Screen

An image-classification model reviews X-ray for visible urgent specialist referral.

Communicate

The summary is read aloud in the patient's own regional language – built for low-literacy and elderly users.



Technologies to be used

- **Frontend:** HTML, CSS, JS PWA (Progressive Web App; works on low-end device; Web app works on Android phones)
- **Backend:** Python (Flask/FastAPI) for OCR and ML requests
- **OCR:** Tesseract OCR (Python) or Google Cloud Vision API
- **NLP simplification & translation:** LLM or Bhashini (bhashini.gov.in – multilingual AI platform)
- **X-ray screening model:** Python + TensorFlow/PyTorch, CNN via transfer
- **Offline-first design:** cache T5 or sync with online APIs
- **Database:** SQLite for the prototype

FEASIBILITY AND VIABILITY

Team
The Persimmons

- **Analysis of the feasibility of the idea**

(Flow Charts/5/imagesed/ working prototype)

- **Technical:** patient photo-reports, medicine terms, and X-rays as meds
- **Operational:** OCR + NLP explanation into plain-nasion-and diagnosis amdan
- **Financial:** Tesseratt positives/negatives
- **Legal:** image-classification X-ray review areas
- **Communicate :** read-aloud summary on the patient

- **Potential challenges and risks**

- Messy handwriting
- Wrong medication info
- X-ray model false positives/negatives
- Limited regional language coverage etc.

- **Strategies for overcoming these challenges**

- MVP focuses on printed reports
- Show original text next to the simplified text next to
- Position as a "flag" tool
- Mai opusis
- Launch with core languages coverage
- Voice-first UI
- On-device processing
- Offline-first design
- Lightweight PWA

IMPACT AND BENEFITS

Potential Impact and Benefits

- **Potential Impact on the target audience**
 - Patients understand their own health (reports, meds)
 - Fewer missed checkups/meds with reminders
 - Reduced anxiety and fear around visits
 - Better outcomes for chronic patients (sugar/BP tracking)
 - Greater independence for elderly and low-literacy users
 - Safer medicine use
 - Builds trust in digital tools (transparent system)
- **Benefits of the solution**
 - Economically : Reduces repeat doctor visits
 - Socially : Improves health literacy
 - Government / Policy : Supports public health goals

- Details / Links of the reference and research work
 - SIH 2026 Problem Statement Portal – sih.gov.in/sih2026PS (PS26133)
 - Bhashini (Government of India's multilingual AI initiative) – bhashini.gov.in
 - NIH Chest X-ray Dataset – kaggle.com/datasets/nih-chest-xrays/data
 - Tesseract OCR – github.com/tesseract-ocr/tesseract
- Quick Note
 - Bhashini → powers the multilingual explanation/translation feature
 - NIH Chest X-ray Dataset → training/reference data for the X-ray flagging
 - Tesseract OCR → handles reading printed/typed reports (the OCR piece from the mitigation slide)
 - SIH portal link → shows which problem statement this submission is addressing